

Mathematical Foundations of Political Methodology

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- Set Theory
- Vectors and Order Lists
- Functions

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- Functions

Set

Definition

The set is a collection of objects viewed as a single entity.

- If the objects in the set can be counted then it's common for the set to be written by enclosing the objects in curly braces $\{\dots\}$
 - The curly braces are extremely important: $\{1, 4, 5\} \neq (1, 4, 5)$
- Examples of sets: $\{\text{Maggie, Bobby}\}$, or $\{2, 5, 3, 8, 10\}$, or $\{\text{red, green, blue}\}$
- The elements of a set must all be unique: $\{1, 1, 3, 4, 4\}$ is not a set (it's a *multiset*)
- The ordering of elements in a set does not matter: $\{\text{red, blue, green}\} = \{\text{green, blue, red}\}$

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Definition

Is an element of is: $\in$.

Examples:

- $\text{green} \in \{\text{red}, \text{green}, \text{blue}\}$
- $\text{magenta} \notin \{\text{red}, \text{green}, \text{blue}\}$

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Subset

Definition

Is a subset of is: $\subset$.

Examples:

- $\{\text{green, red}\} \subset \{\text{red, green, blue}\}$
- Note that $\text{green} \not\subset \{\text{red, green, blue}\}$ because “green” IS NOT a set!
- However $\{\text{green}\} \subset \{\text{red, green, blue}\}$ because $\{\text{green}\}$ IS a set

Remark:

- Sometimes a distinction is made between $\subset$ (strict subset) and $\subseteq$ (weak subset); your book doesn't make the distinction

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Union

Definition

Union is $\cup$, and represents the set of elements in either set.

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- $\{\text{Maggie, Bobby}\} \cup \{2, 5, 3, 8, 10\}$
 $= \{\text{Maggie, Bobby}, 2, 5, 3, 8, 10\}$
- $\{\text{Maggie, Bobby}\} \cup \{2, 5, 3, 8, 10\} \cup \{\text{red, green, blue}\}$
 $= \{\text{Maggie, Bobby}, 2, 5, 3, 8, 10, \text{red, green, blue}\}$
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Intersection

Definition

Intersection is $\cap$, and represents the set of elements in both sets.

Examples:

- $\{\text{Maggie, Bobby}\} \cap \{\text{Bobby, Bogdan}\} = \{\text{Bobby}\}$
- $\{\text{Maggie, Bobby}\} \cap \{2, 5, 3, 8, 10\} = \emptyset$

$\emptyset$ is the “empty set” which is the set that contains nothing, or $\{\}$ The output of union and intersection operations *is always a set!*

- $\{\text{Maggie, Bobby}\} \cap \{\text{Bobby, Bogdan}\} \neq \text{Bobby}$

If two sets have an empty intersection, they are *disjoint*

- $\{\text{Maggie, Bobby}\}$ and $\{2, 5, 3, 8, 10\}$ are disjoint

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Concise Notation for Set

Oftentimes we don't want to (or can't) simply list all the items in a set, so we define sets in other ways (note we *always use the curly braces*)

- The set of all the boys' names that start with a vowel:
 $\{x : x \text{ is a boy's name starting with a vowel}\}$
- The set of all real numbers greater than -17.5 and less than 4:
 $\{x \in \mathbb{R} : -17.5 < x < 4\}$

Using this notation, we can define the following:

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

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Family of Sets and Cardinality of Set

- A *family of sets* is a set whose elements are themselves sets
 - $\{\{1, 2, 4\}, \{2, 6\}, \{\text{green}\}\}$
- A well-known family of sets is the *power set* of any set X : it is the set of all subsets of X , and is written 2^X
 - $2^{\{\text{cat}, \text{dog}\}} = \{\emptyset, \{\text{cat}\}, \{\text{dog}\}, \{\text{cat}, \text{dog}\}\}$
- $|X|$ is the cardinality (size) of set X ; the number of elements in X
 - If $|X| = n$ then $|2^X| = 2^n$

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Set Theory

- If $A \subset B$ then the set of things that are in B but not A is the *complement of A in B*

$$B \setminus A = \text{The complement of } A \text{ in } B = \{x \in B : x \notin A\}$$

- A set with a finite number of elements is a finite set
- A set with an infinite number of elements is an infinite set

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Set of Numbers

- The real numbers, $\mathbb{R}$: All numbers that are positive, negative, or 0
- The natural numbers, $\mathbb{N}$: The set of all strictly positive integers
 $\mathbb{N} = \{1, 2, 3, 4, \dots\}$
- The integers, $\mathbb{Z}$: the natural numbers, their negatives, and 0
 $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$

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Ordered List

- Unlike sets, *ordered lists* are denoted with smooth brackets
(c, a, b, d, e)
 - Elements can repeat
- The 2-dimensional plane is represented by $\mathbb{R}^2$, or $\mathbb{R} \times \mathbb{R}$

$$\mathbb{R}^2 = \{(x, y) : x \in \mathbb{R} \text{ and } y \in \mathbb{R}\}$$

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Cartesian Product

- The general idea of an ordered pair doesn't require it to consist of two real numbers (or even two elements of the same set!)
- Given two sets X and Y , the Cartesian product of X and Y is $\{(x, y) : x \in X \text{ and } y \in Y\}$
 - $\{\text{cat}, \text{dog}\} \times \{a, b\} = \{(\text{cat}, a), (\text{cat}, b), (\text{dog}, a), (\text{dog}, b)\}$
- $X_1 \times X_2 \times \dots \times X_n = \{(x_1, x_2, \dots, x_n) : x_i \in X_i \text{ for } i = 1, 2, \dots, n\}$
- $\mathbb{R}^n$ is the n -ary Cartesian product of $\mathbb{R}$

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Vector

Definition

An ordered list of numbers is a vector.

If the vector has n elements, then it is an element of n -dimensional Euclidean space, or $\mathbb{R}^n$

Examples:

- $(1, 5, 7) \in \mathbb{R}^3$
- $(3, 2, 7, 12, 65, 78787, 3) \in \mathbb{R}^7$

Remark:

- If $n \leq 3$, you can plot out the vector (as either on a line, the plane, or 3-D space). If $n > 3$ you can't plot the vector out, but there is no other conceptual difference

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A function is a mapping from a set X into a set Y

- This means that for each element of X , f assigns an element of Y , or $f : X \rightarrow Y$
- The *graph* of a function from $\mathbb{R}$ to $\mathbb{R}$ is the set

$$\{(x, y) \in \mathbb{R}^2 : y = f(x)\}$$

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- This means that for each element of X , f assigns an element of Y , or $f : X \rightarrow Y$
- The *graph* of a function from $\mathbb{R}$ to $\mathbb{R}$ is the set

$$\{(x, y) \in \mathbb{R}^2 : y = f(x)\}$$

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A function $f : X \rightarrow Y$ is one-to-one (or injective) if every element of X maps into a unique element of Y ($f(a) = f(b) \Rightarrow a = b$)

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A function $f : X \rightarrow Y$ is onto (or surjective) if every element of Y is mapped into by some element of X ($\forall y \in Y, \exists x \in X : f(x) = y$)

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Composite Functions

Definition

Mappings can be combined so that if $f : B \rightarrow C$ and $g : A \rightarrow B$ then $f \circ g = f(g(x))$ is a mapping from A to C : $x \in A$, $g(x) \in B$, and $f(g(x)) \in C$

Examples:

- $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, with $f(x, y) = x + 2y$

- $g : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, with $g(x, y, z) = (x^2 + 2y * z, 7)$

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$$h = f \circ g : \mathbb{R}^3 \rightarrow \mathbb{R}, \text{ with}$$

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In-class problems

3.1.2 Let A, B, C be sets such that $A \subset B$ and $A \cap C = \emptyset$. Simplify the following, if possible:

- $A \cap (B \cup C)$
- $(A \cap B) \cup C$
- $A \cup (B \cap C)$
- $(A \cup B) \cap C$

In-class problems

3.4.1 Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Interpret the mapping $f(x, y) = (x, -y)$ geometrically

- Which points remain unchanged under this mapping?
- Now answer both questions for $f(x, y) = (x, -4)$